

# Bhavya N Jain

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## EDUCATION

|                                 |                                 |          |           |
|---------------------------------|---------------------------------|----------|-----------|
| B.Tech in Computer Science Engg | Vellore Institute of Technology | May 2026 | 8.70 / 10 |
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## SKILLS

**Languages & Frameworks:** Java, Python, JavaScript (ES6+), TypeScript, React, Spring Boot, Node.js, Express

**Tools & Cloud:** AWS, Docker, GitHub Actions, Redis, PostgreSQL, MongoDB, Salesforce, JWT, Git, Agile Methodology

**Concepts:** Scalable Distributed Systems, Test Automation Frameworks, Object-Oriented Programming (OOP), Data Structures & Algorithms, WebSockets, REST APIs, CI/CD, Auth

## TRAINING

**CTDP Trainee — Salesforce Development , PwC India**

Jul '25 – Aug '25

- Mastered CRM configuration across Object Management, App Builder, Security, and Sales & Service Applications.
- Built end-to-end workflow automation skills using Flows, Approval Processes, Data & Analytics, and Salesforce AI.
- Built CRM platform foundation, directly complementing hands-on backend systems engineering in **scalable, distributed apps**.

## PROJECTS

**ByteCache | Core Java 17, OOP, Concurrency, JUnit 5, JMH, Docker, AWS EC2, GitHub Actions**

- Achieved **2.18M** ops/sec (LRU) and **1.07M** ops/sec (LFU) under a **1M-op** quad-threaded load, benchmarked end-to-end via **JMH**.
- Eliminated lock contention via **ReentrantReadWriteLock**, enabling fully parallel reads with serialized writes across **90/10** r/w.
- Engineered a custom **TCP** server with **Redis**-compatible commands (SET/GET/EXPIRE), dual-strategy TTL expiry, and pub/sub.
- Deployed live on **AWS EC2** inside **Docker**, publicly accessible via Telnet/NC for real-world stress testing against a production.
- Built an automated test framework using JUnit 5 and JMH, instrumenting throughput/latency metrics to validate performance under concurrent load.

**Pathfinder Duel | Java 21, Spring Boot 3, WebSockets (STOMP), Redis, PostgreSQL, React 18 + TypeScript**

- Built a real-time **2-player** strategy game on a shrinking **11x11** grid with sub-**50ms** move latency via STOMP WebSockets.
- Implemented a **BFS** wall-validation engine that rejects any wall isolating a player in **O(n)** time, guaranteeing **100%** game-state.
- Designed dual-storage architecture: **Redis** for hot game state + **PostgreSQL** for persistent match history, decoupling low-latency.
- Engineered an automated test framework (JUnit 5 + GitHub Actions CI) gating every push with backend test coverage, TypeScript checks, and build validation.

**DriveEasy | MERN Stack, React, Node.js, Express, MongoDB, JWT, bcrypt, ImageKit CDN, Vercel**

- Shipped a production full-stack marketplace with dual owner/customer roles, real-time availability, and a responsive dashboard.
- Prevented double-bookings via server-side date-conflict validation at the DB layer, protecting booking integrity.
- Integrated ImageKit CDN to improve render performance across all vehicle listing pages without backend code changes.
- Secured all routes with **JWT + bcrypt** and **Role-based Access Control**, achieving zero plaintext credential exposure and auth for admin CRUD operations.

## ACHIEVEMENTS

- Competitive Programming: maintained a **145+** day consecutive coding streak and solved **200+** problems across LeetCode and GeeksforGeeks, consistently reinforcing strong **Data Structures, Algorithms**, and problem-solving fundamentals.
- Actively participated in multiple **hackathons**, demonstrating collaborative problem-solving and rapid prototyping skills.

## EXTRACURRICULAR ACTIVITIES

**GirlScript Summer of Code (GSSoC) 2026**

Present

- Ranked Top 1% among 43,000+ contributors, collaborating with maintainers across 18+ open-source repositories to resolve production-grade issues including XSS vulnerability patches, CI/CD pipeline failures, and DB transaction consistency bugs.
- Consistently shipped **security**-hardening PRs and automated test suites across varied stacks including **React, Node.js**, and **Python** projects, demonstrating collaborative engineering rigour and reliable cross-team delivery at open-source scale.

## CERTIFICATIONS

- AWS Academy Graduate Cloud Architecting & Cloud Foundations:** high-availability system design, IAM policy management, scalable multi-tier deployments, and cloud security fundamentals.
- IBM – GEN AI** Using Watsonx: prompt engineering and hands-on AI solution building on the Watsonx platform.